length-that is, when R takes on the value 1—the winding and vibration modes have comparable mass. The two approaches to measuring distance become equally difficult to carry out and, moreover, each would yield the same result since 1 is its own reciprocal.

As the radius continues to shrink, the winding modes become lighter than the unwound modes and hence, since we are always opting for the "easier approach," they should now be used to measure distances. According to this method of measurement, which yields the reciprocal of that measured by the unwound modes, the radius is larger than one times the Planck length and increasing. This simply reflects that as R—the quantity measured by unwound strings—shrinks to 1 and continues to get smaller, 1/R—the quantity measured by wound strings—grows to 1 and gets larger. Therefore, if one takes care to always use the light string modes—the "easy" approach to measuring distance—the minimal value encountered is the Planck length.

In particular, a big crunch to zero size is avoided, as the radius of the universe as measured using light string-mode probes is always larger than the Planck length. Rather than heading through the Planck length on to ever smaller size, the radius, as measured by the lightest string modes, decreases to the Planck length and then immediately starts to increase. The crunch is replaced by a bounce.

Using light string modes to measure distances aligns with our conventional notion of length—the one that was around long before the discovery of string theory. It is according to this notion of distance, as seen in Chapter 5, that we encountered insurmountable problems with violent quantum undulations if sub-Planck-scale distances play a physical role. We once again see, from this complementary perspective, that the ultrashort distances are avoided by string theory. In the physical framework of general relativity and in the corresponding mathematical framework of Riemannian geometry there is a single concept of distance, and it can acquire arbitrarily small values. In the physical framework of string theory, and, correspondingly, in the realm of the emerging discipline of quantum geometry, there are two notions of distance. By judiciously making use of both we find a concept of distance that meshes with both our intuition and with general relativity when distance scales are large, but